problem: I aimed to analyze cryptocurrency market trends using data visualization techniques.

how cryptocurrency prices fluctuate over time

the relationship between price and trading volume

correlations between different financial metrics

market share of top cryptocurrencies

using bokeh i created interctive visualizations to help analysts understand patterns in crypto market effectively.

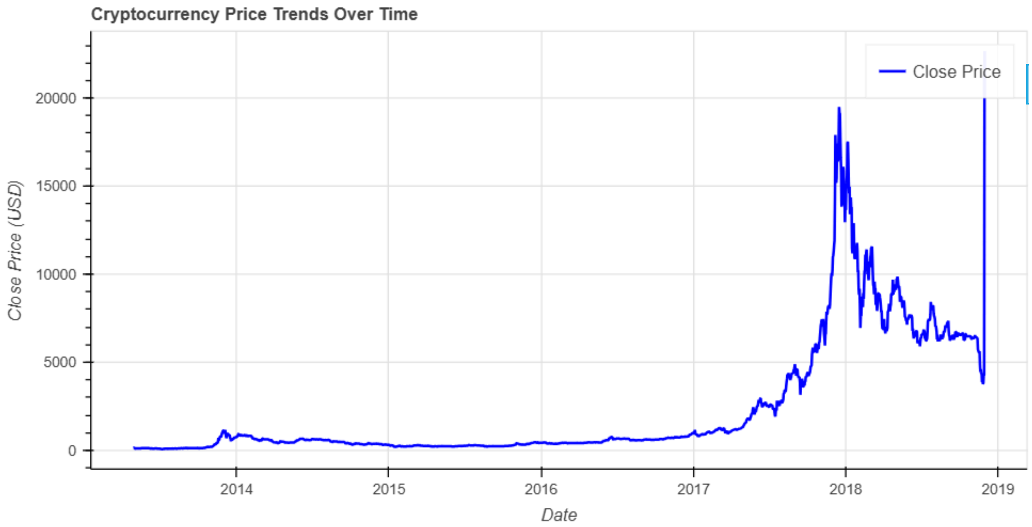
This project presents a detailed analysis of cryptocurrency market trends using Bokeh. Through interactive visualizations, we explore price movements, trading volumes, and market share distributions to gain insights into market behavior. The dataset includes historical price data, transaction volumes, and ranking trends, helping to identify key patterns over time

```
import pandas as pd
import numpy as np

# Load your dataset (replace with actual path or link)
df = pd.read_csv("/content/crypto-markets.csv")

# Step 1: Convert 'date' column to datetime format
df['date'] = pd.to_datetime(df['date'])

# Step 2: Check for missing values
missing_values = df.isnull().sum()
print("Missing Values:", missing_values)

# Step 3: Drop rows with missing values in essential columns (if needed)
df = df.dropna(subset=['date', 'open', 'high', 'low', 'close']).copy()


# Step 4: Check the data types to ensure prices are numeric
df['open'] = pd.to_numeric(df['open'], errors='coerce')
df['high'] = pd.to_numeric(df['high'], errors='coerce')
df['low'] = pd.to_numeric(df['low'], errors='coerce')
df['close'] = pd.to_numeric(df['close'], errors='coerce')

# Step 5: Drop duplicates if there are any
df = df.drop_duplicates(subset=['date'])

# Step 6: Final check
print(df.head())
print(df.dtypes)

df = df.sort_values('date')
df = df.drop_duplicates(subset=['date'])
df = df.sort_values('date')  # Ensure chronological order
```

```
Missing Values: slug          0
symbol         0
name           0
date           0
ranknow        0
open           0
high           0
low            0
close          0
volume         0
market         0
close_ratio    0
spread         0
dtype: int64
      slug symbol     name       date  ranknow    open    high     low  \
0  bitcoin    BTC  Bitcoin 2013-04-28        1  135.30  135.98  132.10
1  bitcoin    BTC  Bitcoin 2013-04-29        1  134.44  147.49  134.00
2  bitcoin    BTC  Bitcoin 2013-04-30        1  144.00  146.93  134.05
3  bitcoin    BTC  Bitcoin 2013-05-01        1  139.00  139.89  107.72
4  bitcoin    BTC  Bitcoin 2013-05-02        1  116.38  125.60   92.28

    close  volume        market  close_ratio  spread
0  134.21     0.0  1.488567e+09       0.5438    3.88
1  144.54     0.0  1.603769e+09       0.7813   13.49
2  139.00     0.0  1.542813e+09       0.3843   12.88
3  116.99     0.0  1.298955e+09       0.2882   32.17
4  105.21     0.0  1.168517e+09       0.3881   33.32
slug                   object
symbol                 object
name                   object
date           datetime64[ns]
ranknow                 int64
```

```
open                  float64
high                  float64
low                   float64
close                 float64
volume                float64
market                float64
close_ratio           float64
spread                float64
dtype: object
```

```
from bokeh.plotting import figure, show, output_notebook
import pandas as pd

# Output the plot in the notebook
output_notebook()

# Creating the Bokeh figure
p = figure(width=800, height=400, x_axis_type="datetime", title="Cryptocurrency Price Trends Over Time")

# Adding a line renderer with legend and line thickness
p.line(df['date'], df['close'], legend_label="Close Price", line_width=2, color="blue")

# Adding labels
p.xaxis.axis_label = 'Date'
p.yaxis.axis_label = 'Close Price (USD)'

# Show the plot
show(p)
```



Description: This chart visualizes the closing price of cryptocurrency over time, highlighting market fluctuations.

Insights:

Prices remained stable until 2016, followed by a sharp rise in 2017.

A sudden peak and decline indicate market volatility and corrections.

```
import pandas as pd
from bokeh.plotting import figure, show, output_notebook
from bokeh.models import HoverTool, ColumnDataSource
from bokeh.io import output_file

output_notebook()

# Load dataset
df = pd.read_csv( "/content/crypto-markets.csv", parse_dates=['date'])

# Group by week to reduce noise
df_weekly = df.groupby(pd.Grouper(key='date', freq='W'))['close'].mean().reset_index()

# Convert to Bokeh data source
source = ColumnDataSource(df_weekly)

# Create figure
p = figure(title="Cryptocurrency Closing Price Trends", x_axis_label='Date', y_axis_label='Closing Price',
           x_axis_type='datetime', y_axis_type='log', width=900, height=500)
```

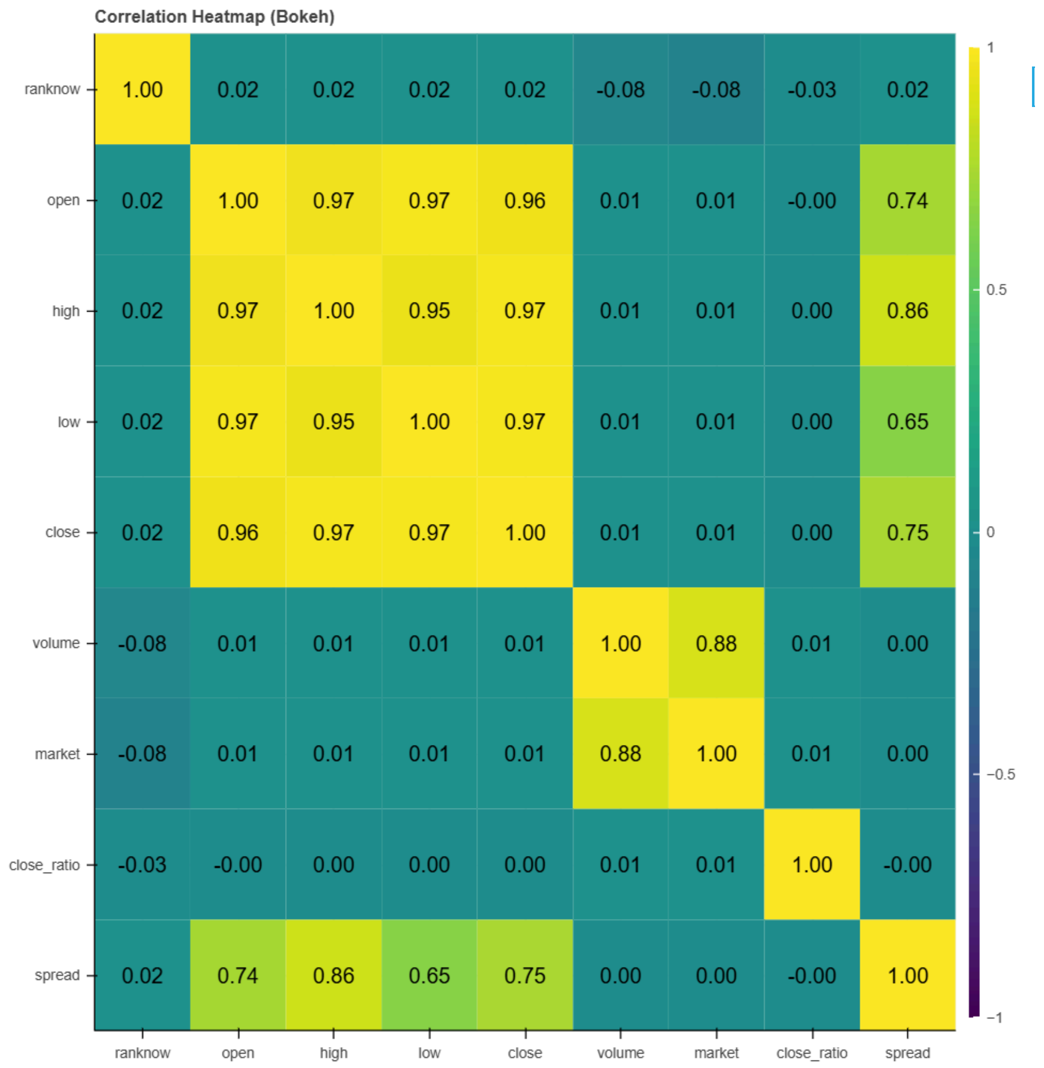
```
# Add area (fill_between equivalent in Bokeh)
p.varea(x='date', y1=1e-5, y2='close', source=source, fill_color="skyblue", fill_alpha=0.5)

# Add line plot
p.line('date', 'close', source=source, line_width=2, line_color="blue", legend_label="Close Price")

# Add hover tool
hover = HoverTool(tooltips=[("Date", "@date{%F}"), ("Close", "@close{0.0000}")], formatters={'@date': 'datetime'})
p.add_tools(hover)

# Improve layout
p.legend.location = "top_right"
p.legend.click_policy = "hide"
p.grid.grid_line_alpha = 0.3

# Show plot
show(p)
```

Description: The area plot illustrates the fluctuation of cryptocurrency closing prices over time, emphasizing market trends.

Insights: The market remained relatively stable initially, followed by a sharp surge and a sudden drop, indicating high volatility.

```
from bokeh.plotting import figure, show
from bokeh.io import output_notebook
from bokeh.models import ColumnDataSource, LinearColorMapper, ColorBar, LabelSet
import pandas as pd
import numpy as np

output_notebook()

# Load dataset
df = pd.read_csv("crypto-markets.csv")

# Drop non-numeric columns
df_numeric = df.select_dtypes(include=[np.number])

# Compute correlation matrix
corr_matrix = df_numeric.corr()

# Convert matrix into long format for Bokeh
corr_matrix_long = corr_matrix.stack().reset_index()
corr_matrix_long.columns = ['Variable1', 'Variable2', 'Correlation']

# Prepare data for Bokeh
source = ColumnDataSource(corr_matrix_long)

# Define color mapper (Red-Blue for better contrast)
mapper = LinearColorMapper(palette="Viridis256", low=-1, high=1)

# Create figure
p = figure(title="Correlation Heatmap (Bokeh)",
           x_range=list(df_numeric.columns),
           y_range=list(reversed(df_numeric.columns)),
```

```
           width=800, height=800,
           tools="pan,box_zoom,reset,save")

# Add rectangles (heatmap cells)
p.rect(x="Variable1", y="Variable2", width=1, height=1, source=source,
       line_color=None, fill_color={'field': 'Correlation', 'transform': mapper})

# Add correlation values as text labels inside the boxes
source_labels = ColumnDataSource(data={
    'x': corr_matrix_long['Variable1'],
    'y': corr_matrix_long['Variable2'],
    'text': [f"{val:.2f}" for val in corr_matrix_long['Correlation']]
})
labels = LabelSet(x='x', y='y', text='text', source=source_labels,
                  text_align="center", text_baseline="middle", text_color="black")

p.add_layout(labels)

# Add color bar
color_bar = ColorBar(color_mapper=mapper, width=8, location=(0,0))
p.add_layout(color_bar, 'right')

# Show plot
show(p)
```

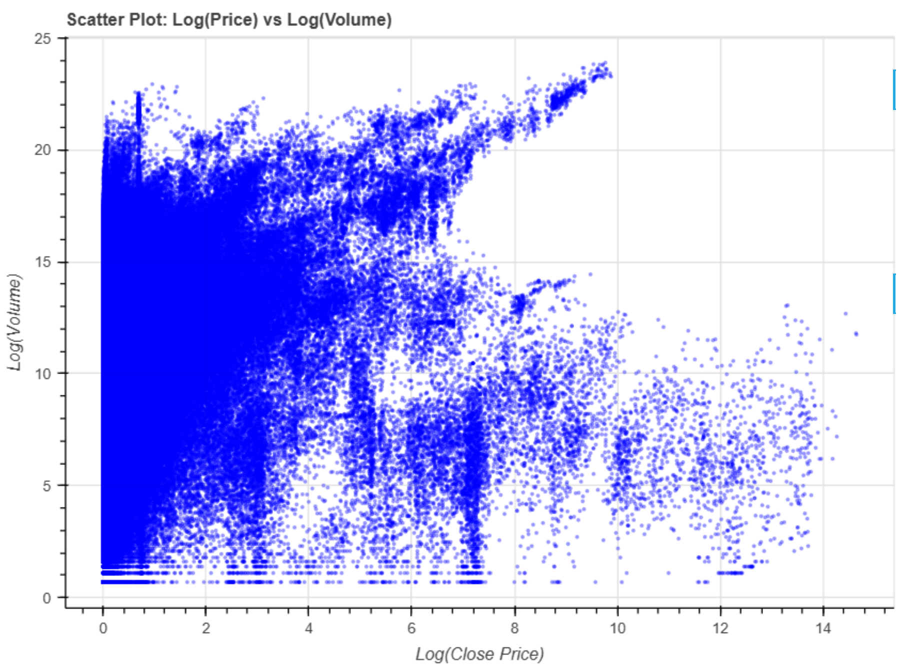


Description: This heatmap visualizes the correlation between different cryptocurrency attributes, with color intensity indicating the strength of relationships.

Insights: Strong positive correlations are observed among open, high, low, and close prices, suggesting that these features move together. Volume and market capitalization show weaker correlations with price metrics.

```
import pandas as pd
import numpy as np
from bokeh.plotting import figure, show, output_notebook
from bokeh.models import HoverTool
```

```
# Load data
df = pd.read_csv("crypto-markets.csv")

# Remove rows where volume is 0 and close price is too small
df = df[(df["volume"] > 0) & (df["close"] > 0.001)].copy()

# Apply log transformation with log1p (handles log(0) safely)
df["log_close"] = np.log1p(df["close"])
df["log_volume"] = np.log1p(df["volume"])

# Enable Bokeh in Colab
output_notebook()

# Create Bokeh scatter plot
p = figure(title="Scatter Plot: Log(Price) vs Log(Volume)",
           x_axis_label="Log(Close Price)", y_axis_label="Log(Volume)",
           tools="pan,wheel_zoom,box_zoom,reset,save", width=700, height=500)

# Add scatter points
p.scatter(df["log_close"], df["log_volume"], color="blue", alpha=0.3, size=2)

# Add interactive hover tool
hover = HoverTool(tooltips=[("Log Price", "@x"), ("Log Volume", "@y")])
p.add_tools(hover)

# Show plot
show(p)
```

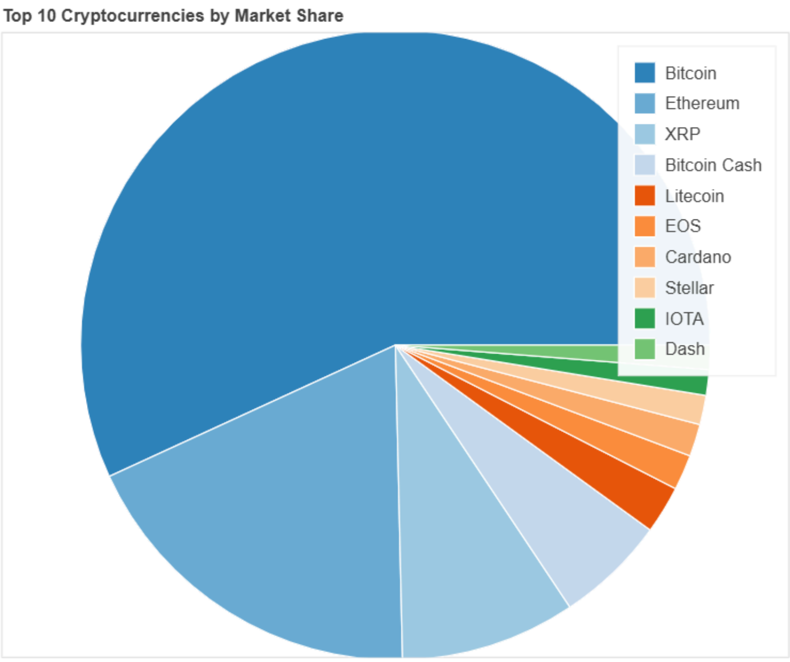


Description: This scatter plot represents the relationship between closing prices and trading volumes of cryptocurrencies. Each point corresponds to a transaction, plotted on a logarithmic scale for better visibility.

Insights: A wide spread of points indicates high variance in trading activity. No strong linear relationship is observed, suggesting that price movements are not directly proportional to trading volume.

```
from bokeh.plotting import figure, show
from bokeh.transform import cumsum
from bokeh.palettes import Category20c
from math import pi
import pandas as pd

# Load dataset
df = pd.read_csv("/content/crypto-markets.csv")

# Aggregate market share of each cryptocurrency
df_grouped = df.groupby('name')['market'].sum().reset_index()
df_grouped = df_grouped.sort_values(by='market', ascending=False)

# Select top 10 cryptocurrencies by market share
df_top10 = df_grouped.head(10).copy()

# Calculate market share percentage
df_top10['market_share'] = df_top10['market'] / df_top10['market'].sum()
```

```
# Calculate angles for pie chart
df_top10['angle'] = df_top10['market_share'] * 2 * pi

# Assign colors
df_top10['colors'] = Category20c[len(df_top10)]

# Create figure (Fixed height issue)
p = figure(height=500, title="Top 10 Cryptocurrencies by Market Share",
           toolbar_location=None, tools="")

# Add wedges to the pie chart
p.wedge(x=0, y=0, radius=0.8,
        start_angle=cumsum('angle', include_zero=True),
        end_angle=cumsum('angle'),
        line_color="white", fill_color='colors',
        legend_field='name', source=df_top10)

# Remove axis and grid
p.axis.axis_label = None
p.axis.visible = False
p.grid.grid_line_color = None

# Show plot
show(p)
```



Description: This pie chart illustrates the market share distribution among the top 10 cryptocurrencies based on market capitalization. Each segment represents a cryptocurrency, with the size proportional to its market dominance.

Insights: A few major cryptocurrencies hold a significant portion of the market, while others have comparatively smaller shares. This highlights the concentration of market value among leading digital assets.